Credit Risk Assessment

US-901 Retrieve Credit Assessment Inputs

User Story

As the Loan Origination System,

I want to retrieve all validated customer, financial, collateral, and legal information required for credit risk assessment,

So that a comprehensive credit risk evaluation can be performed before making a lending decision.

Acceptance Criteria

Scenario 1: Successful Retrieval

Given the customer has completed the Financial Analysis and Collateral Assessment,

When the Credit Risk Assessment process begins,

Then

- Customer Profile shall be retrieved.
- Financial Analysis results shall be retrieved.
- Collateral Assessment results shall be retrieved.
- Legal Due Diligence results shall be retrieved.
- Existing Loan Exposure shall be retrieved.
- Credit Bureau Report shall be retrieved.
- Internal Risk Rating shall be retrieved.
- Banking Relationship History shall be retrieved.
- The retrieved information shall be linked to the corresponding Loan Application ID.

Scenario 2: Missing Information

Given the Credit Risk Assessment process has started,

When mandatory information is unavailable,

Then

- The missing information shall be identified.
- The application shall be flagged.

- The application shall not proceed until all mandatory information is available.

Business Rules

- Only validated Financial Analysis and Collateral Assessment results shall be used.
- All retrieved information shall be linked to the Loan Application ID.
- Missing mandatory information shall prevent Credit Risk Assessment.

US-902 Calculate Credit Risk Parameters

User Story

As the Loan Origination System,

I want to calculate key credit risk parameters using the customer's financial, collateral, and credit information,

So that objective credit risk indicators are available for AI evaluation and credit decision-making.

Acceptance Criteria

Scenario 1: Successful Calculation

Given all required information is available,

When Credit Risk Parameters are calculated,

Then

The system shall calculate:

- Debt Service Coverage Ratio (DSCR)
- Debt-to-Income Ratio (DTI)
- Loan-to-Value Ratio (LTV)
- Current Ratio
- Interest Coverage Ratio
- Existing Debt Exposure
- Cash Flow Adequacy
- Repayment Capacity
- Probability of Default Indicators

Scenario 2: Calculation Failure

Given mandatory financial information is incomplete,

When parameter calculation is attempted,

Then

- The missing parameters shall be identified.
- Credit Risk Assessment shall not proceed until required information is available.

Business Rules

- Risk parameters shall be calculated using validated financial information.
- All calculations shall comply with the bank's credit policy.

US-903 Perform AI Credit Risk Assessment

User Story

As the AI Credit Risk Assessment Engine,

I want to evaluate the customer's credit risk using financial analysis, collateral assessment, legal due diligence, and credit risk parameters,

So that the Credit Officer receives an AI-assisted credit risk recommendation to support lending decisions.

Acceptance Criteria

Scenario 1: Successful AI Assessment

Given all required assessment inputs are available,

When the AI Credit Risk Assessment is executed,

Then

The AI shall evaluate:

- Repayment Capacity
- Probability of Default
- Financial Strength
- Industry Risk
- Business Stability
- Credit Behaviour
- Cash Flow Trend
- Portfolio Concentration
- Overall Risk Rating

Scenario 2: Insufficient Data

Given mandatory assessment information is unavailable,

When AI Credit Risk Assessment is executed,

Then

- The assessment shall be flagged as incomplete.
- The application shall proceed for manual review.

Business Rules

- AI recommendations are advisory only.
- AI shall not replace the Credit Officer's decision.

US-904 AI Evaluation & Governance

User Story

As the AI Evaluation & Governance Engine,

I want to evaluate the trustworthiness of the AI-generated Credit Risk Assessment,

So that only reliable and explainable AI recommendations are used during the credit appraisal process.

Acceptance Criteria

Scenario 1: Governance Evaluation

Given the AI Credit Risk Assessment has been completed,

When governance evaluation is performed,

Then

The system shall evaluate:

- AI Confidence Score
- Explainability
- Bias Detection
- Data Quality Validation
- Hallucination Detection
- Policy Compliance

Scenario 2: Governance Passed

Given all governance checks are successful,

When evaluation is completed,

Then

- The AI recommendation shall be marked as Governance Approved.
- The application shall proceed to Credit Decision.

Scenario 3: Governance Failed

Given governance exceptions are detected,

When evaluation is completed,

Then

- The application shall be routed for manual review.
- Governance exceptions shall be recorded.

Business Rules

- Governance evaluation shall be mandatory before AI recommendations are used.
- Failed governance checks shall prevent automated workflow progression.

US-905 Review Credit Risk Assessment & Recommendation

User Story

As the Loan Origination System,

I want to automatically route loan applications to the Credit Officer whenever AI governance exceptions, business policy thresholds, or delegated authority limits are triggered,

So that the Credit Officer can perform an independent assessment, document observations, recommend appropriate risk mitigation measures, and submit the application to the Sanctioning Authority for a final lending decision.

Acceptance Criteria

Scenario 1: AI Governance Exception

Given the AI Credit Risk Assessment has been completed,

When any of the following conditions are detected:

- AI Confidence Score is below the configured threshold.
- AI Bias is detected.

- Explainability is unavailable.
- Hallucination is detected.
- Critical financial information is missing.
- Governance validation fails.

Then

- The application shall be routed to the Credit Officer.
- Governance exceptions shall be recorded.
- Automated loan approval shall not proceed.

Scenario 2: High Credit Risk

Given the AI Credit Risk Assessment has been completed,

When

- Credit Risk Rating = High, or
- Probability of Default exceeds the bank's acceptable threshold,

Then

- The application shall be assigned to the Credit Officer.
- A detailed manual risk assessment shall be performed.
- Additional mitigants may be requested before proceeding.

Scenario 3: Delegated Authority Limit Exceeded

Given the loan application has been assessed,

When

- The requested loan amount exceeds the Credit Officer's delegated authority limit,

Then

- The application shall be reviewed by the Credit Officer.
- The Credit Officer shall provide a recommendation.
- The application shall be forwarded to the appropriate Sanctioning Authority.

Scenario 4: Policy Exception

Given the credit assessment has been completed,

When any lending policy exception is identified, including but not limited to:

- Loan-to-Value (LTV) exceeds the approved threshold.

- Industry exposure exceeds concentration limits.
- Collateral coverage is below the minimum requirement.
- Customer credit policy deviation is identified.
- Regulatory exception exists.

Then

- The application shall be routed for manual review.
- The Credit Officer shall document the policy exception.
- Appropriate risk mitigation measures shall be recommended.

Scenario 5: Credit Officer Review

Given the application has been assigned,

When the Credit Officer completes the assessment,

Then the Credit Officer may:

- Recommend Approval
- Recommend Rejection
- Request Additional Information
- Recommend additional collateral/security
- Recommend revised loan amount
- Recommend revised repayment terms

The recommendation shall include supporting comments and justification.

Business Rules

- AI recommendations are advisory and shall not replace human judgment.
- Manual review shall be mandatory whenever governance or business thresholds are exceeded.
- Credit Officer recommendations shall be documented before submission to the Sanctioning Authority.
- All recommendations shall be recorded in the audit trail.

US-906 Approve or Reject Loan by Sanctioning Authority

User Story

As the Sanctioning Authority,

I want to review the complete credit assessment package, including the Credit Officer's recommendation, AI assessment, governance evaluation, financial analysis, collateral assessment, and legal due diligence,

So that I can make an informed lending decision in accordance with the bank's credit policy and delegated approval authority.

Acceptance Criteria

Scenario 1: Loan Approved

Given the application has been recommended for approval,

When the Sanctioning Authority approves the application,

Then

- The approval decision shall be recorded.
- Approved loan amount shall be stored.
- Approved terms and conditions shall be stored.
- The application shall proceed to Loan Documentation.

Scenario 2: Loan Rejected

Given the application has been reviewed,

When the Sanctioning Authority rejects the application,

Then

- The rejection reason shall be recorded.
- The applicant status shall be updated.
- The application shall be closed.

Scenario 3: Request Additional Information

Given additional clarification is required,

When the Sanctioning Authority requests further information,

Then

- The application shall return to the Credit Officer.
- Required information shall be recorded.
- Processing shall resume after the requested information is received.

Scenario 4: Conditional Approval

Given the application is acceptable subject to additional conditions,

When the Sanctioning Authority issues a conditional approval,

Then

Conditions may include:

- Additional collateral
- Additional guarantor
- Lower sanctioned amount
- Higher margin contribution
- Additional financial documentation
- Enhanced monitoring

The conditions shall be recorded before loan documentation begins.

Business Rules

- Only authorized Sanctioning Authorities may approve loans.
- Approval authority shall be determined by the bank's Delegation of Financial Powers (DoFP) matrix.
- Approval decisions shall comply with regulatory and internal credit policies.
- All approvals and rejections shall be fully auditable.

US-907 Store Credit Risk Assessment Results

User Story

As the Loan Origination System,

I want to securely store the complete Credit Risk Assessment, including AI recommendations, governance evaluation results, Credit Officer recommendations, Sanctioning Authority decisions, and audit information,

So that the lending decision is fully traceable, auditable, and available throughout the loan lifecycle.

Acceptance Criteria

Scenario 1: Successful Storage

Given the lending decision has been finalized,

When the application is completed,

Then the system shall store:

- Credit Risk Rating
- Probability of Default
- Credit Score
- AI Credit Risk Assessment
- AI Governance Evaluation Result
- Credit Officer Recommendation
- Sanctioning Authority Decision
- Approved Loan Amount
- Approved Terms and Conditions
- Review Comments
- Assessment Date & Time
- Loan Application ID

Scenario 2: Audit Trail

Given the assessment has been stored,

When the transaction is completed,

Then the audit log shall capture:

- Assessment Date & Time
- AI Model Version
- Confidence Score
- Governance Status
- Credit Officer ID
- Sanctioning Authority ID
- Approval/Rejection Decision
- User/System ID
- Version History

Business Rules

- Only finalized decisions shall be stored.
- All records shall be linked to the corresponding Loan Application ID.
- Assessment records shall be immutable after final approval.
- Audit logs shall comply with the bank's record retention policy.

- Role-based access control shall be enforced for all stored assessment records.

Could not identify user